

MAC 233: Hall Effect Sensor Lab

Dr Dan Brennan

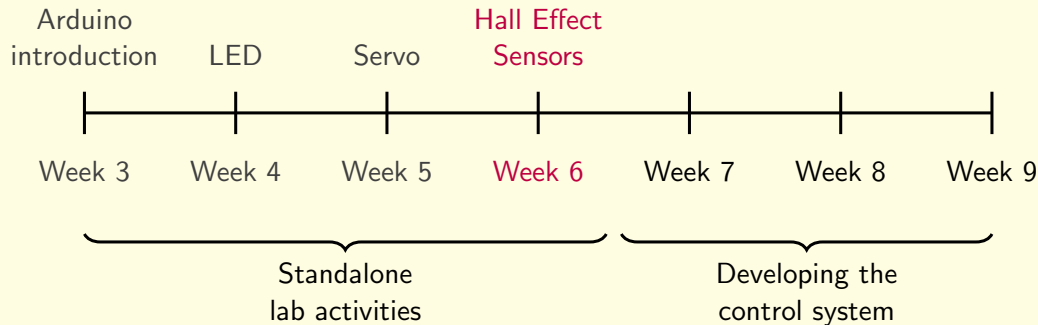
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MAC 233: **05/11/2025**



University of
Sheffield

MAC 233: LABS OVERVIEW



TODAY'S LAB: PREREQUISITES

This lab doesn't require any code from previous labs; however, it does expect you to have completed these, as common code repeated from previous labs will not be explained in detail again here.

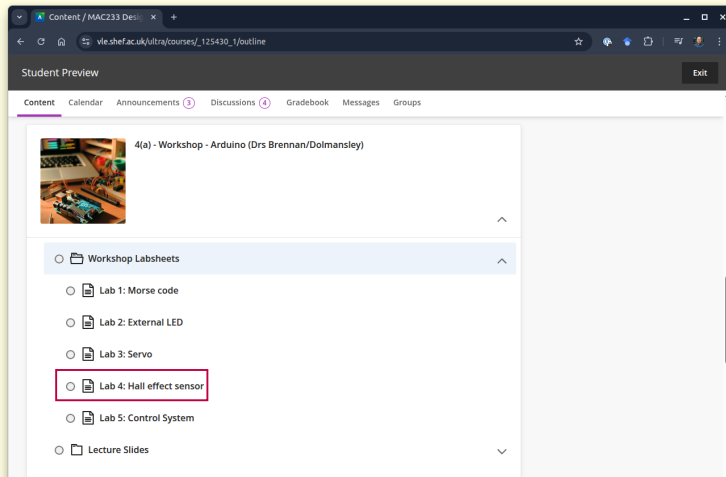
Out of your Arduino pack, you will need: 1 Arduino, 1 A3144 hall effect sensor, 1 LED, 1 $10K\Omega$ resistor, 1 220Ω resistor, and 5 jumper cables.

TODAY'S LAB: OBJECTIVES

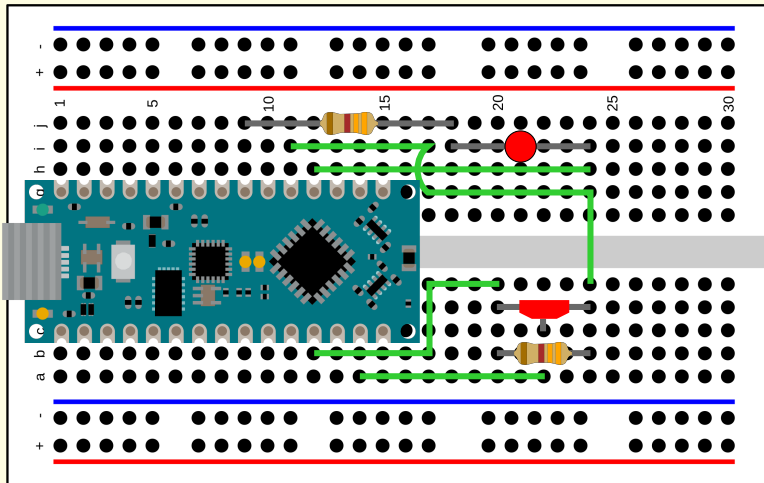
The objective of today's lab is to learn how to use an interrupt to record a value of a sensor whose value is changed via an external event: <https://docs.arduino.cc/language-reference/en/functions/external-interrupts/attachInterrupt/>.

LAB 4: HALL EFFECT

Login to Blackboard: <https://vle.shef.ac.uk/>



WIRING DIAGRAM



Breadboard source: <https://freesvg.org/breadboard>, Resistor source <https://freesvg.org/resistor-330-ohm>.